



BIOLOGY CH.20 — DISCOVERIES & VACCINES — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

Topics in this chapter:

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5. Other Vaccines
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1. Gregor Mendel & Laws of Inheritance

- Gregor Johann Mendel ne 3 laws of inheritance diye hain. Unko "Father of Genetics" kaha jata hai. 1856-1863 tak pea plants par experiments kiye. [Q1139, Q1141, Q1142, Q1158, Q1162]
- Mendel ki 3 laws: - Law of Dominance: hybrid offspring dominant trait inherit karti hai - Law of Segregation (Purity of Gametes): hereditary factors ki 2 copies segregate hoti hain; offspring ko har parent se ek factor milta hai - Law of Independent Assortment: traits ke pairs independently segregate hote hain [Q1139, Q1161]
- Mendel ki success ka main reason: numerical count of progeny rakhi - science aur mathematics ka blend kiya. Large sample size, pea plants (distinct traits), aur several generations tak inferences confirm kiye. [Q1141, Q1142]
- Law of Segregation (purity of gametes) monohybrid cross ke results se derive kiya gaya tha. [Q1161]
- Darwin = Evolution theory. Watson & Crick = DNA structure. Carl Linnaeus = Taxonomy. [Q1139, Q1158]
- Law of Independent Assortment: traits ke pairs independently segregate hote hain [Q1139, Q1161]

2. COVID-19 Vaccines & Drugs

- Vincov-19: India ki first indigenous COVID-19 drug. Immunised horses ke blood serum ko synthesise karke banayi gayi. [Q1140]
- DCGI approved 2 indigenous vaccines: COVISHIELD (Serum Institute of India) aur COVAXIN (Bharat Biotech + ICMR-National Institute of Virology). [Q1140, Q1152]
- ZyCoV-D: India's first DNA plasmid vaccine. Sputnik V = Russia ka (Gamaleya Research Institute). [Q1140]
- COVAXIN (BBV152) = whole inactivated virus-based vaccine. Bharat Biotech HQ = Genome Valley, Hyderabad. ICMR HQ = New Delhi. [Q1152]

3. Scientific Discoveries & Scientists

- Binomial nomenclature (genus + species, e.g., *Homo sapiens*) = Carolus Linnaeus ne introduce kiya. Father of Taxonomy. [Q1143, Q1159]
- Crescograph (plant growth measure karne ka instrument) = Sir Jagadish Chandra Bose ne invent kiya. [Q1145]
- Theophrastus = Father of Botany. Aristotle = Father of Zoology. Antoine Lavoisier = Father of Chemistry. [Q1146]
- Smallpox vaccine = Edward Jenner ne 1796 mein Varicella virus use karke invent kiya. [Q1147]
- Blood groups (A, B, AB, O) = Karl Landsteiner ne discover kiye. Universal donor = O negative, Universal recipient = AB positive. [Q1149]
- Protoplasm term = Johann Evangelist Purkinje ne coin kiya. Protoplasm = cytoplasm + nucleus + organelles (plasma membrane se surrounded). [Q1150]
- Cell = Robert Hooke ne 1665 mein cork observe karke discover kiya. Antony Van Leeuwenhoek ne pehle living cells discover kiye. Cytology = cell ka study. [Q1151]
- Penicillin (first antibiotic) = Alexander Fleming ne 1928 mein *Penicillium notatum* fungus se discover kiya. [Q1153]
- James Chadwick = neutron discover kiya (1932). Dmitri Mendeleev = Modern Periodic Table. John Dalton = Atomic Theory. Isaac Newton = Laws of Gravity and Motion. [Q1147, Q1149, Q1153]

4. Antibiotics & Medicines

- Antibiotics = bacterial infection treat karne ke liye. Bacteria ko kill/inhibit karte hain. [Q1144]
- Anesthetic = sensation loss induce karta hai. Antiseptic = living tissues (wounds, cuts) par infection prevent karta hai. Antipyretic = fever reduce karta hai. [Q1144]
- Penicillin ke alawa: Azithromycin, Erythromycin, Streptomycin. [Q1153]

5. Other Vaccines

- ZikaVac (Zika virus vaccine) = Bharat Biotech (Hyderabad, India) ne develop kiya. Zika virus Aedes mosquito ke bite se spread hota hai. [Q1154]
- Rabies vaccine = Louis Pasteur ne develop kiya. [Q1156]
- ERVEBO = Ebola vaccine (Merck). [Q1154]

6. Medical Techniques & Instruments

- Ultrasonics = foetal growth monitor karne ki latest technique. Obstetrics/gynaecology mein use hoti hai. [Q1155]
- CAT Scan = 3D images ke liye X-rays use karta hai. Eudiometer = gas mixture ki volume change measure karta hai. Chronometer = time measure. Bolometer = radiation measure. Cardiograph = heart rate measure. [Q1145, Q1155]

7. Fermentation & Pasteur

- Louis Pasteur (Father of Bacteriology) ne fermentation process, pasteurization, aur anthrax/rabies vaccines develop kiye. French chemist aur microbiologist. [Q1156, Q1157]
- Robert Koch ne tuberculosis bacterium, cholera, aur anthrax discover kiye. Edward Jenner ne smallpox vaccine. [Q1156, Q1157]

8. Taxonomy & Classification

- Linnaean hierarchy (broadest to specific): Kingdom → Phylum → Class → Order → Family → Genus → Species. [Q1163]
- Taxonomy = naming, describing, classifying organisms based on behavioral, genetic, biochemical variations. [Q1159]

9. Evolution

- "Survival of the fittest" phrase = Herbert Spencer ne coin kiya. [Q1148]
- Charles Darwin ne Theory of Evolution by Natural Selection propose kiya - book 'On the Origin of Species'. Life simple se complex forms mein evolve hui. [Q1148, Q1160]
- Darwin's theory: individuals with survival/reproduction characteristics un traits ko offspring mein pass karte hain. [Q1160]
- Albert Einstein = Theory of Relativity. Archimedes = Law of Buoyancy. [Q1148]
- J.B.S. Haldane aur Aleksandr Oparin = chemical evolution theory. Stanley L. Miller = Father of prebiotic chemistry. [Q1158]

10. ⚠ Traps / Key Differences

- Darwin = Evolution, Mendel = Inheritance. [Q1139, Q1158]
- AIDS = disease, HIV = virus. [Q1081]
- Ascending order = smallest to largest; lekin Q1163 mein "ascending" ke baad bhi correct answer broad-to-specific (Kingdom→Species) hai - ye exam ka tricky part hai. [Q1163]
- Mendel = monohybrid cross se Law of Segregation derive kiya. [Q1161]